



10145 - A tale of two disks: Chemical characterisation of the planet-hosting protoplanetary disk and a physical characterisation of the circumplanetary disk

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	WISPIT-2	MIRI Medium Resolution Spectroscopy	(1) WISPIT-2
	2	WISPIT-2 bkg	MIRI Medium Resolution Spectroscopy	(2) WISPIT-2-bkg
	3	WISPIT-2	MIRI Medium Resolution Spectroscopy	(1) WISPIT-2
	4	WISPIT-2 bkg	MIRI Medium Resolution Spectroscopy	(2) WISPIT-2-bkg
	5	WISPIT-2	MIRI Medium Resolution Spectroscopy	(1) WISPIT-2
	6	WISPIT-2 bkg	MIRI Medium Resolution Spectroscopy	(2) WISPIT-2-bkg

ABSTRACT

JWST Proposal 10145 (Created: Friday, March 13, 2026, 2:00:19PM Eastern Standard Time) - Overview

The recent discovery of the embedded protoplanet WISPIT 2b makes WISPIT 2 the first system where we may test typical planet formation in action. Given the 0.32" separation of the planet from its host star, it is the only system of its kind where we can study both the terrestrial planet-forming zone around WISPIT 2A, and the presence and geometry of a circumplanetary disk (CPD) around WISPIT 2b with MIRI/MRS. We expect the cold dust of the CPD to be especially bright in MIRI/MRS. We can fit the CPD's temperature structure to constrain the leading accretion mechanism, which significantly impact the derived accretion rate of WISPIT 2b, and will provide essential information for planetary accretion models.

Furthermore, Jupiter is said to have blocked a part of the outer disk material from replenishing the inner disk with water, thereby influencing the composition of Earth. From PDS 70 it is known that, despite the presence of two planets within its cavity, material is still reaching the inner disk. In comparison, WISPIT 2 - with the protoplanet instead located in a gap - represents a more typical case given the high occurrence rate of disks hosting gaps and rings. It therefore provides the perfect laboratory to test how Jupiter may have influenced the terrestrial planet-forming region of our own Solar System. We therefore propose to observe the WISPIT 2 system with MIRI/MRS in order to: (i) detect the thermal emission of the protoplanet WISPIT 2b to confirm the presence of a CPD and characterise its physical properties; and (ii) obtain the spectrum of the circumstellar disk to perform chemical and physical characterisation of dust and gas in the inner regions.

OBSERVING DESCRIPTION

We request observations of WISPIT 2 and dedicated sky backgrounds. The observations of WISPIT 2 are set up such that we can extract a spectrum from both the protoplanet WISPIT 2b and the primary's inner disk. We place WISPIT 2 on the nominal 4-point dither pattern positions with Target Acquisition using the FND filter. By doing so, we can use optimal data reduction methods to extract the highest quality spectrum from the primary. Furthermore, by limiting the PA to 100-109.6deg, we place the companion as far as possible from the primary on the detector, whilst avoiding the MOZ and maintaining schedulability. The large physical distance in the detector-space enables us to optimally remove the PSF on the detector, and use optimal detector-based extraction methods to get the highest quality spectrum of the companion. Using 100 groups and 4 integrations, we are able to go deep while avoiding negative effects from cosmic rays. We predict to get an $S/N > 500$ on the primary, and around 14.5 on the companion. The $S/N > 500$ on the primary ensures that we will be able to detect the faintest of molecular features. The ideal S/N of 14.5 gives sufficient leeway to reach our science goal of detecting the presence of a circumplanetary disk around the companion, and its physical structure.

Proposal 10145 - Targets - A tale of two disks: Chemical characterisation of the planet-hosting protoplanetary disk and a physical char...

	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
Fixed Targets	(1)	WISPIT-2	RA: 19 23 17.0332 (290.8209717d) Dec: -07 40 55.08 (-7.68197d) Equinox: J2000	Proper Motion RA: 6.308 mas/yr Proper Motion Dec: -27.137999904880417 mas/yr Parallax: 0.0074649" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Protoplanetary disks, Young stellar objects] Extended=NO				
Fixed Targets	(2)	WISPIT-2-bkg	RA: 19 23 19.6151 (290.8317296d) Dec: -07 40 59.25 (-7.68313d) Equinox: J2000	Proper Motion RA: 6.308 mas/yr Proper Motion Dec: -27.137999904880417 mas/yr Parallax: 0.0074649" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Protoplanetary disks, Young stellar objects] Extended=NO				

Proposal 10145 - Observation 1 - A tale of two disks: Chemical characterisation of the planet-hosting protoplanetary disk and a physic...

Observation	Proposal 10145, Observation 1: WISPIT-2												Fri Mar 13 19:00:20 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[WISPIT-2 bkg (Obs 2)]												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	WISPIT-2	RA: 19 23 17.0332 (290.8209717d) Dec: -07 40 55.08 (-7.68197d) Equinox: J2000				Proper Motion RA: 6.308 mas/yr Proper Motion Dec: -27.137999904880417 mas/yr Parallax: 0.0074649" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star												
	Description=[Protoplanetary disks, Young stellar objects] Extended=NO												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	FND	FAST	4	1	1	11.1	268233				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction		
	All MRS		NO				FULL				Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SHORT(A)	MRSLONG		FASTR1	100	4	1	Dither 1	4	16	4473.364	
	1	SHORT(A)	MRSSHORT		FASTR1	100	4	1	Dither 1	4	16	4473.364	

Proposal 10145 - Observation 1 - A tale of two disks: Chemical characterisation of the planet-hosting protoplanetary disk and a physic...

Special Requirements	Aperture PA Range 100 to 109.6 Degrees (V3 100.0 to 109.6) 3 After 1 by 0 Days to 6 Days Sequence Observations 2, 1 (reordered), Non-interruptible
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Proposal 10145 - Observation 2 - A tale of two disks: Chemical characterisation of the planet-hosting protoplanetary disk and a physic...

Observation	Proposal 10145, Observation 2: WISPIT-2 bkg												Fri Mar 13 19:00:20 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [WISPIT-2 (Obs 1)]												
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(2)	WISPIT-2-bkg	RA: 19 23 19.6151 (290.8317296d) Dec: -07 40 59.25 (-7.68313d) Equinox: J2000				Proper Motion RA: 6.308 mas/yr Proper Motion Dec: -27.137999904880417 mas/yr Parallax: 0.0074649" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star												
	Description=[Protoplanetary disks, Young stellar objects]												
Acquisition	Extended=NO												
	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	FND	All MRS				NO			FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	2-Point				BACKGROUND				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SHORT(A)	MRSLONG		FASTR1	100	4	1	Dither 1	2	8	2236.682	
	1	SHORT(A)	MRSSHORT		FASTR1	100	4	1	Dither 1	2	8	2236.682	

Special Requirements	Sequence Observations 2, 1 (reordered), Non-interruptible
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Proposal 10145 - Observation 3 - A tale of two disks: Chemical characterisation of the planet-hosting protoplanetary disk and a physic...

Observation	Proposal 10145, Observation 3: WISPIT-2												Fri Mar 13 19:00:20 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[WISPIT-2 bkg (Obs 4)]												
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	WISPIT-2	RA: 19 23 17.0332 (290.8209717d) Dec: -07 40 55.08 (-7.68197d) Equinox: J2000				Proper Motion RA: 6.308 mas/yr Proper Motion Dec: -27.137999904880417 mas/yr Parallax: 0.0074649" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star												
	Description=[Protoplanetary disks, Young stellar objects] Extended=NO												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	Optional ETC ID			
	1	SAME	FND	FAST	4	1		1	11.1	268233			
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction		
	All MRS		NO				FULL				Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	MEDIUM(B)	MRSLONG		FASTR1	100	4	1	Dither 1	4	16	4473.364	
	1	MEDIUM(B)	MRSSHORT		FASTR1	100	4	1	Dither 1	4	16	4473.364	

Proposal 10145 - Observation 3 - A tale of two disks: Chemical characterisation of the planet-hosting protoplanetary disk and a physic...

Special Requirements	Aperture PA Range 100 to 109.6 Degrees (V3 100.0 to 109.6) 3 After 1 by 0 Days to 6 Days 5 After 3 by 0 Days to 6 Days Sequence Observations 4, 3 (reordered), Non-interruptible
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Proposal 10145 - Observation 4 - A tale of two disks: Chemical characterisation of the planet-hosting protoplanetary disk and a physic...

Observation	Proposal 10145, Observation 4: WISPIT-2 bkg												Fri Mar 13 19:00:20 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [WISPIT-2 (Obs 3)]												
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(2)	WISPIT-2-bkg	RA: 19 23 19.6151 (290.8317296d) Dec: -07 40 59.25 (-7.68313d) Equinox: J2000				Proper Motion RA: 6.308 mas/yr Proper Motion Dec: -27.137999904880417 mas/yr Parallax: 0.0074649" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Protoplanetary disks, Young stellar objects] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	FND	All MRS				NO			FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	2-Point				BACKGROUND				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	MEDIUM(B)	MRSLONG		FASTR1	100	4	1	Dither 1	2	8	2236.682	
	1	MEDIUM(B)	MRSSHORT		FASTR1	100	4	1	Dither 1	2	8	2236.682	

Special Requirements	Sequence Observations 4, 3 (reordered), Non-interruptible
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Proposal 10145 - Observation 5 - A tale of two disks: Chemical characterisation of the planet-hosting protoplanetary disk and a physic...

Observation	Proposal 10145, Observation 5: WISPIT-2												Fri Mar 13 19:00:20 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[WISPIT-2 bkg (Obs 6)]												
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	WISPIT-2	RA: 19 23 17.0332 (290.8209717d) Dec: -07 40 55.08 (-7.68197d) Equinox: J2000				Proper Motion RA: 6.308 mas/yr Proper Motion Dec: -27.137999904880417 mas/yr Parallax: 0.0074649" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star												
	Description=[Protoplanetary disks, Young stellar objects] Extended=NO												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	FND	FAST	4	1	1	11.1	268233				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction		
	All MRS		NO				FULL				Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	LONG(C)	MRSLONG		FASTR1	100	4	1	Dither 1	4	16	4473.364	
	1	LONG(C)	MRSSHORT		FASTR1	100	4	1	Dither 1	4	16	4473.364	

Proposal 10145 - Observation 5 - A tale of two disks: Chemical characterisation of the planet-hosting protoplanetary disk and a physic...

Special Requirements	Aperture PA Range 100 to 109.6 Degrees (V3 100.0 to 109.6) 5 After 3 by 0 Days to 6 Days Sequence Observations 6, 5 (reordered), Non-interruptible
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Proposal 10145 - Observation 6 - A tale of two disks: Chemical characterisation of the planet-hosting protoplanetary disk and a physic...

Observation	Proposal 10145, Observation 6: WISPIT-2 bkg												Fri Mar 13 19:00:20 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [WISPIT-2 (Obs 5)]												
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(2)	WISPIT-2-bkg	RA: 19 23 19.6151 (290.8317296d) Dec: -07 40 59.25 (-7.68313d) Equinox: J2000				Proper Motion RA: 6.308 mas/yr Proper Motion Dec: -27.137999904880417 mas/yr Parallax: 0.0074649" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star												
	Description=[Protoplanetary disks, Young stellar objects] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	FND	All MRS				NO			FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	2-Point				BACKGROUND				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	LONG(C)	MRSLONG		FASTR1	100	4	1	Dither 1	2	8	2236.682	
	1	LONG(C)	MRSSHORT		FASTR1	100	4	1	Dither 1	2	8	2236.682	

Special Requirements	Sequence Observations 6, 5 (reordered), Non-interruptible
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